

Name: _____

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MyDesign Portfolio

Element K Initial Template

Element K: Reflection on the Design Project

Engineering Design Process Summary (K1, K2)

Summarize each step of the design process including:

- defining the problem (elements A-C)

We identified the problem of not enough shelving for 3D printer filaments after discussion with Mr. L and other stakeholders; recorded previous attempts to solve it, the attempts in his room were haphazardly retrofitted and failed to solve the issue; and wrote out high-priority design requirements that would be necessary for a good final design, such as the ability to store all of the filament rolls and sufficiently organize them.

- ideation (element D)

We brainstormed ideas, discussed them between ourselves, and sketched them out and described them. We came up with around five viable ideas, though it was clear that only two of them were really competing with each other for the top spot, and were fairly similar.

- selection & prototype design (elements D-F)

We put the ideas into a scoring matrix to select which was best. The highest score design, and the one we selected, was the modular shelving unit design, hanging horizontally and vertically. We designed our idea according to STEM principles and considered its risks and its viability. We fully detailed the selected design, modeled it in CAD, and then listed all the materials and costs of those materials for the design. We then created a plan of construction in a Gantt chart.

- prototype building (element G)

We built and iterated on the shelf prototype to better meet our requirements. We used cardboard, glue, and x-acto knives to quickly and cheaply assemble the prototype. The final prototype ended up being a portion of the total modular shelving unit, which worked well because of the modular nature of the design, and we could test everything we needed to test.

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- testing (elements H-I)

We created a very detailed testing plan for the prototype, tested it, analyzed the data, and interpreted the results. The testing all went well, and the prototype passed on all fronts, meeting the four high-priority design requirements.

- evaluation (element J)

We got an evaluation of the prototype from multiple stakeholders and a field expert. The evaluations were very positive, with no gaps or issues in the prototype assessed. Mr. L suggested changing one of the materials for it to be lighter and cheaper.

For each project step, explain: What went well? What didn't go well? Why?

This is also sort of a summary

- defining the problem (elements A-C)

These steps all went very well, we were able to clearly define the problem after talking with Mr. L and other stakeholders. This was sort of a niche issue, so there weren't so many examples of prior solutions we could find online though. We were able to create clear and easily measurable design requirements. These steps went well, especially the first one, because of thorough planning and thinking about which questions to ask.

- ideation (element D)

This step didn't go as well. We were too focused on evaluating the specifics and details of certain ideas rather than truly brainstorming many, also our ideas were generally pretty similar to each other. However, we did generate good ideas.

- selection & prototype design (elements D-F)

These steps went fairly well, the prototype selected through the scoring matrix was generally agreed upon to be the best, and we were able to create a good and specific design modeled in CAD. The design was viable and based on STEM principles. This was due to good planning and thinking through every aspect of the design before putting it in.

- prototype building (element G)

For our first barebones cardboard prototype, everything went well, and it met all of the testing requirements. There were significant issues after that, though. We planned to create a fully-functioning wood prototype, but we didn't have enough time at the end, and we could not complete it. So, while testing was good for the



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rudimentary prototype, we couldn't test a fully fledged prototype at all. We ran into this issue because we were too slow on the earlier steps, and should have taken less time on those and instead focused on building the prototype.

- testing (elements H-I)

This step went well. We were able to create robust tests for each design requirement, allowing us to fully evaluate the prototypes. This was mentioned a bit above, but for the first prototype which we were able to test passed with flying colors and testing went very smoothly. This step went well due to creating very solid design requirements earlier, which were easy to test.

- evaluation (element J)

This step went very well. We got good evaluations on the prototype from Mr. L and student stakeholders, and the prototype lined up with the recommendations from the field expert at NOVA Fab Labs. The field expert part didn't go as well, as we had limited correspondence and not as much feedback as from the stakeholders.

Lessons Learned About the Engineering Design Process (K3)

What do you think designers need to consider or do based on your experience? A level 5 reflection illustrates thorough and specific details on improving the design process in the future. Make sure to touch on lessons from across the whole engineering design process and experience that can apply to any future engineer. The lessons learned in Element K should focus on the engineering design process and how you implemented it. Later on (in L1) you will provide recommendations for continuing work on this project in particular.

Based on our experience, designers should strongly consider the time they spend on each step of the engineering process, and make sure to allocate enough time to the construction of prototypes rather than just thorough planning. Also, they should make sure to fully detail the prototype in CAD. This will help with the time as well, as they can accurately assess how long it will take to build the prototype and all of its intricacies.